Jiaxun Li

jasonli@umich.edu jiaxun-li.github.io

Statistics PhD student Statistical inference, learning theory, and simulation research.
Seeking Summer 2027 quantitative research internships.

Education

University of Michigan, Ann Arbor Sep. 2024 – expected Sep. 2029
Ph.D. in Statistics Advisors: Ambuj Tewari and Saptarshi Chakraborty
Selected coursework: Probability Theory, Statistical Inference, Monte Carlo Methods, Advanced Statistical Learning, Computational Optimization, Information Theory and Statistics.

Xi'an Jiaotong University Sep. 2020 – Aug. 2024
B.Sc. in Mathematics, Honors Program GPA: 4.21/4.30

Selected Research and Publications

Exact and conservative inference for the average treatment effect in stratified experiments with binary outcomes *Under review, 2025*
Jiaxun Li, Jacob Spertus, Philip B. Stark

- Implemented simulation experiments for statistical inference in stratified randomized experiments with binary outcomes, connecting experimental design and causal inference.

Generation through the lens of learning theory *COLT 2025*
Jiaxun Li*, Vinod Raman*, Ambuj Tewari *Equal contribution

- Developed mathematical frameworks and theoretical characterizations for generation; studied its relationship with prediction and online learnability.
- Contributed presentation at COLT 2025.

On generation in metric spaces *NeurIPS 2026, forthcoming*
Jiaxun Li, Vinod Raman, Ambuj Tewari

- Used complexity measures and mathematical proofs to study generative capabilities in metric spaces and their dependence on geometry and novelty scales.

Robust Detection of LLM-Generated Text under Contamination *Under review, 2026*
Jiaxun Li, Saptarshi Chakraborty, Ambuj Tewari

- Independently designed a benchmark and directed Python experiments for robust detection, evaluating AUROC and performance at low false-positive rates.
- Implemented models and experimental pipelines, and personally directed the analysis.

Technical Skills

Programming: Python, R, MATLAB, C/C++
Libraries: NumPy, SciPy, PyTorch, scikit-learn **Tools:** Bash, Git, LaTeX

Teaching and Recognition

Graduate Student Instructor, University of Michigan 2024 – 2026
Taught Probability and Distribution Theory and undergraduate probability and statistics.
Outstanding Graduate Student Instructor Team Award, STATS 485 (Winter 2026).